Lesson 2: Measures and the Carath eodory Extension

Curriculum: A Rigorous Learning Path to Multi-Agent Deep RL

February 9, 2026

Contents

3 Measures **2**

3.1 Definition and Fundamental Properties 2

3.2 Null Sets and Complete Measures 3

4 Construction of Measures: Carathéodory's Theorem **3**

4.1 Algebras and Pre-Measures 3

4.2 Outer Measures 3

4.3 Carathéodory's Extension Theorem 4

4.4 Application: Lebesgue Measure 5

3 Measures

3.1 Definition and Fundamental Properties

Definition 3.1 (Measure)**.** A *measure* on a measurable space $(\Omega, \mathcal{F})$ is a function $\mu : \mathcal{F} \to [0, \infty]$ satisfying:

(i) $\mu(\emptyset) = 0$.

(ii) **Countable additivity:** If $A_1, A_2, \ldots \in \mathcal{F}$ are pairwise disjoint, then:

$$\mu\left(\bigsqcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu(A_n). \tag{1}$$

The triple $(\Omega, \mathcal{F}, \mu)$ is a *measure space*. If $\mu(\Omega) = 1$, then μ is a *probability measure* and we write $\mathbb{P}$ instead of μ.

Example 3.1 (Standard Measures)**.** (a) **Counting measure:** $\mu(A) = |A|$ (number of elements) on any countable Ω.

(b) **Dirac measure:** $\delta_x(A) = \mathbf{1}_A(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A \end{cases}$.

(c) **Lebesgue measure:** $\lambda((a, b]) = b - a$ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. (Existence proved in Section 4.)

(d) **Probability on a finite set:** $\mathbb{P}(\{s\}) = p(s)$ with $\sum_{s \in \mathcal{S}} p(s) = 1$.

Theorem 3.1 (Properties of Measures)**.** *Let $(\Omega, \mathcal{F}, \mu)$ be a measure space. For $A, B, A_n \in \mathcal{F}$:*

(a) ***Monotonicity:*** $A \subseteq B \implies \mu(A) \leq \mu(B)$.

(b) ***Subtractivity:*** $A \subseteq B$ *and* $\mu(A) < \infty \implies \mu(B \setminus A) = \mu(B) - \mu(A)$.

(c) ***Sub-additivity:*** $\mu(\bigcup_{n=1}^{\infty} A_n) \leq \sum_{n=1}^{\infty} \mu(A_n)$.

(d) ***Continuity from below:*** $A_n \uparrow A \implies \mu(A) = \lim_n \mu(A_n)$.

(e) ***Continuity from above:*** $A_n \downarrow A$ *and* $\mu(A_1) < \infty \implies \mu(A) = \lim_n \mu(A_n)$.

Proof. **(a)** $B = A \sqcup (B \setminus A)$, so $\mu(B) = \mu(A) + \mu(B \setminus A) \geq \mu(A)$.

(b) From $\mu(B) = \mu(A) + \mu(B \setminus A)$ and $\mu(A) < \infty$, subtract $\mu(A)$.

(c) Define $B_1 = A_1$, $B_n = A_n \setminus \bigcup_{k=1}^{n-1} A_k$ for $n \geq 2$. The B_n are disjoint, $\bigcup B_n = \bigcup A_n$, and $B_n \subseteq A_n$. So: $\mu(\bigcup A_n) = \mu(\bigsqcup B_n) = \sum \mu(B_n) \leq \sum \mu(A_n)$.

(d) Let $A_0 = \emptyset$. The sets $C_n = A_n \setminus A_{n-1}$ are disjoint and $A_n = \bigsqcup_{k=1}^{n} C_k$. Also $A = \bigsqcup_{k=1}^{\infty} C_k$. By countable additivity:

$$\mu(A) = \sum_{k=1}^{\infty} \mu(C_k) = \lim_{n \to \infty} \sum_{k=1}^{n} \mu(C_k) = \lim_{n \to \infty} \mu(A_n). \tag{2}$$

(e) Define $B_n = A_1 \setminus A_n$. Then $B_n \uparrow A_1 \setminus A$. By (d): $\mu(A_1 \setminus A) = \lim \mu(B_n) = \lim(\mu(A_1) - \mu(A_n))$, using (b) (valid since $\mu(A_n) \leq \mu(A_1) < \infty$). So $\mu(A_1) - \mu(A) = \mu(A_1) - \lim \mu(A_n)$, giving $\mu(A) = \lim \mu(A_n)$. □

Finiteness Condition in (e)

The condition $\mu(A_1) < \infty$ in continuity from above is essential. Consider counting measure on $\mathbb{N}$ with $A_n = \{n, n+1, n+2, \ldots\}$. Then $A_n \downarrow \emptyset$ but $\mu(A_n) = \infty$ for all n, while $\mu(\emptyset) = 0$. For probability measures, this condition is always satisfied since $\mathbb{P}(A_1) \leq 1 < \infty$.

3.2 Null Sets and Complete Measures

Definition 3.2 (Null Set, Complete Measure)**.** A set $N \in \mathcal{F}$ is a *μ-null set* (or *null set*) if $\mu(N) = 0$. A measure μ is *complete* if whenever N is μ-null and $A \subseteq N$, then $A \in \mathcal{F}$ (and hence $\mu(A) = 0$).

Proposition 3.2 (Completion of a Measure)**.** *Every measure space $(\Omega, \mathcal{F}, \mu)$ can be completed: there exists a smallest σ-algebra $\overline{\mathcal{F}} \supseteq \mathcal{F}$ and a measure $\bar{\mu}$ extending μ such that $(\Omega, \overline{\mathcal{F}}, \bar{\mu})$ is complete. Explicitly:*

$$\overline{\mathcal{F}} = \{A \cup N' : A \in \mathcal{F},\ N' \subseteq N \text{ for some } \mu\text{-null } N \in \mathcal{F}\}, \tag{3}$$

and $\bar{\mu}(A \cup N') = \mu(A)$.

Proof. We verify that $\overline{\mathcal{F}}$ is a σ-algebra. The key step is closure under complement: if $E = A \cup N'$ with $N' \subseteq N$ and $\mu(N) = 0$, then $E^c = A^c \cap (N')^c = (A^c \cap N^c) \cup (A^c \cap N \cap (N')^c)$. The first part is in $\mathcal{F}$, and the second is a subset of N (a null set), so $E^c \in \overline{\mathcal{F}}$.

For well-definedness of $\bar{\mu}$: if $A_1 \cup N_1' = A_2 \cup N_2'$, then $A_1 \triangle A_2 \subseteq N_1 \cup N_2$ (a null set), so $\mu(A_1 \triangle A_2) = 0$, hence $\mu(A_1) = \mu(A_2)$. □

4 Construction of Measures: Carathéodory's Theorem

How do we actually *build* measures? We cannot simply assign $\mu(A)$ for every $A \in \mathcal{B}(\mathbb{R})$ individually — the Borel σ-algebra is uncountably large. Instead, we specify μ on a small, manageable collection and extend it.

4.1 Algebras and Pre-Measures

Definition 4.1 (Algebra of Sets)**.** A collection $\mathcal{A} \subseteq \mathcal{P}(\Omega)$ is an *algebra* (or *field*) if:

(i) $\Omega \in \mathcal{A}$.
(ii) $A \in \mathcal{A} \implies A^c \in \mathcal{A}$.
(iii) $A, B \in \mathcal{A} \implies A \cup B \in \mathcal{A}$. (closed under *finite* union)

Remark 4.1. An algebra is like a σ-algebra but with only finite (not countable) closure. Every σ-algebra is an algebra, but not conversely.

Example 4.1. On $\mathbb{R}$, the collection $\mathcal{A}$ of all finite unions of half-open intervals $(a, b]$ (including $\emptyset$ and $\mathbb{R} = (-\infty, \infty)$) is an algebra but not a σ-algebra. For instance, $\bigcup_{n=1}^{\infty}(0, 1 - 1/n] = (0, 1)$ is not a finite union of half-open intervals, so $(0, 1) \notin \mathcal{A}$.

Definition 4.2 (Pre-Measure)**.** A *pre-measure* on an algebra $\mathcal{A}$ is a function $\mu_0 : \mathcal{A} \to [0, \infty]$ satisfying:

(i) $\mu_0(\emptyset) = 0$.
(ii) If $A_1, A_2, \ldots \in \mathcal{A}$ are disjoint and $\bigsqcup_{n=1}^{\infty} A_n \in \mathcal{A}$, then $\mu_0(\bigsqcup A_n) = \sum \mu_0(A_n)$.

4.2 Outer Measures

Definition 4.3 (Outer Measure)**.** An *outer measure* on Ω is a function $\mu^* : \mathcal{P}(\Omega) \to [0, \infty]$ satisfying:

(i) $\mu^*(\emptyset) = 0$.
(ii) **Monotonicity:** $A \subseteq B \implies \mu^*(A) \leq \mu^*(B)$.
(iii) **Countable sub-additivity:** $\mu^*(\bigcup_{n=1}^{\infty} A_n) \leq \sum_{n=1}^{\infty} \mu^*(A_n)$.

Proposition 4.1 (Pre-Measure Induces Outer Measure)**.** *Given a pre-measure μ_0 on an algebra $\mathcal{A}$, define for any $E \subseteq \Omega$:*

$$\mu^*(E) = \inf\left\{\sum_{n=1}^{\infty} \mu_0(A_n) : A_n \in \mathcal{A},\ E \subseteq \bigcup_{n=1}^{\infty} A_n\right\}. \tag{4}$$

Then μ^* is an outer measure, and $\mu^*(A) = \mu_0(A)$ for all $A \in \mathcal{A}$.

Proof. *(i)* $\mu^*(\emptyset) = 0$: cover $\emptyset$ by $A_n = \emptyset$ for all n.

(ii) If $A \subseteq B$, every cover of B is also a cover of A, so the infimum over covers of A is at most the infimum over covers of B.

(iii) Let $E = \bigcup_n E_n$. For each $\varepsilon > 0$ and each n, choose a cover $\{A_{n,k}\}_k \subseteq \mathcal{A}$ with $E_n \subseteq \bigcup_k A_{n,k}$ and $\sum_k \mu_0(A_{n,k}) \leq \mu^*(E_n) + \varepsilon/2^n$. Then $\{A_{n,k}\}_{n,k}$ covers E, so:

$$\mu^*(E) \leq \sum_n \sum_k \mu_0(A_{n,k}) \leq \sum_n (\mu^*(E_n) + \varepsilon/2^n) = \sum_n \mu^*(E_n) + \varepsilon. \tag{5}$$

Since ε is arbitrary, $\mu^*(E) \leq \sum_n \mu^*(E_n)$.

μ^* extends μ_0: For $A \in \mathcal{A}$, the single-set cover gives $\mu^*(A) \leq \mu_0(A)$. For the reverse: if $A \subseteq \bigcup_n A_n$ with $A_n \in \mathcal{A}$, define $B_n = A \cap A_n \cap (\bigcup_{k<n} A_k)^c \in \mathcal{A}$ (since $\mathcal{A}$ is an algebra). Then $A = \bigsqcup B_n$ and $B_n \subseteq A_n$. By the pre-measure properties: $\mu_0(A) = \sum \mu_0(B_n) \leq \sum \mu_0(A_n)$. Taking the infimum: $\mu_0(A) \leq \mu^*(A)$. □

4.3 Carathéodory's Extension Theorem

Definition 4.4 (Carathéodory Measurability)**.** Given an outer measure μ^* on Ω, a set $A \subseteq \Omega$ is *μ^*-measurable* (or *Carathéodory measurable*) if for every $E \subseteq \Omega$:

$$\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c). \tag{6}$$

Remark 4.2. By sub-additivity, $\mu^*(E) \leq \mu^*(E \cap A) + \mu^*(E \cap A^c)$ always holds. So the content of (6) is the reverse inequality: A "splits" every test set E additively. This is a strong condition — it must hold for *all* $E \subseteq \Omega$, not just sets in some σ-algebra.

Carathéodory's Extension Theorem

Theorem 4.2 (Carathéodory)**.** *Let μ^* be an outer measure on Ω. Then:*

(a) The collection $\mathcal{M} = \{A \subseteq \Omega : A \text{ is } \mu^*\text{-measurable}\}$ is a σ-algebra.

(b) The restriction $\mu = \mu^*|_{\mathcal{M}}$ is a complete measure on $(\Omega, \mathcal{M})$.

Consequently, if μ_0 is a pre-measure on an algebra $\mathcal{A}$ and μ^* is the induced outer measure (4), then $\mathcal{A} \subseteq \mathcal{M}$ and $\mu|_{\mathcal{A}} = \mu_0$. If additionally μ_0 is σ-finite, then μ is the unique extension of μ_0 to $\sigma(\mathcal{A})$.

Proof. **Part (a): $\mathcal{M}$ is a σ-algebra.**

Step 1: $\Omega \in \mathcal{M}$. For any E: $\mu^*(E \cap \Omega) + \mu^*(E \cap \emptyset) = \mu^*(E) + 0 = \mu^*(E)$. ✓

Step 2: Closure under complement. The condition (6) is symmetric in A and A^c, so $A \in \mathcal{M} \Leftrightarrow A^c \in \mathcal{M}$. ✓

Step 3: Closure under finite union. Let $A, B \in \mathcal{M}$. For any test set E:

$$\begin{aligned}\mu^*(E) &= \mu^*(E \cap A) + \mu^*(E \cap A^c) && (A \text{ measurable})\\ &= \mu^*(E \cap A) + \mu^*(E \cap A^c \cap B) + \mu^*(E \cap A^c \cap B^c) && (B \text{ measurable, applied to } E \cap A^c)\end{aligned}$$

We need $\mu^*(E) = \mu^*(E \cap (A \cup B)) + \mu^*(E \cap (A \cup B)^c)$.

Note $(A \cup B)^c = A^c \cap B^c$, so the third term is correct. For the first: $E \cap (A \cup B) = (E \cap A) \cup (E \cap A^c \cap B)$, so by sub-additivity: $\mu^*(E \cap (A \cup B)) \leq \mu^*(E \cap A) + \mu^*(E \cap A^c \cap B)$.

Combining: $\mu^*(E) \geq \mu^*(E \cap (A \cup B)) + \mu^*(E \cap (A \cup B)^c)$. The reverse holds by sub-additivity. So $A \cup B \in \mathcal{M}$. ✓

Step 4: Closure under countable disjoint union. Let $A_1, A_2, \ldots \in \mathcal{M}$ be disjoint. Set $S_n = \bigsqcup_{k=1}^n A_k$ and $S = \bigsqcup_{k=1}^\infty A_k$.

By induction (using Step 3 and closure under complement, hence closure under intersection), $S_n \in \mathcal{M}$, and for any E:

$$\mu^*(E \cap S_n) = \sum_{k=1}^n \mu^*(E \cap A_k). \tag{7}$$

This is proved by induction: $\mu^*(E \cap S_n) = \mu^*(E \cap S_n \cap A_n) + \mu^*(E \cap S_n \cap A_n^c) = \mu^*(E \cap A_n) + \mu^*(E \cap S_{n-1})$.

Since $S_n \in \mathcal{M}$: $\mu^*(E) = \mu^*(E \cap S_n) + \mu^*(E \cap S_n^c)$. Since $S_n \subseteq S$, $S^c \subseteq S_n^c$, so $\mu^*(E \cap S_n^c) \geq \mu^*(E \cap S^c)$. Using (7):

$$\mu^*(E) \geq \sum_{k=1}^n \mu^*(E \cap A_k) + \mu^*(E \cap S^c). \tag{8}$$

Letting $n \to \infty$:

$$\mu^*(E) \geq \sum_{k=1}^\infty \mu^*(E \cap A_k) + \mu^*(E \cap S^c) \geq \mu^*(E \cap S) + \mu^*(E \cap S^c), \tag{9}$$

where the last step uses sub-additivity $\mu^*(E \cap S) \leq \sum \mu^*(E \cap A_k)$. The reverse inequality holds by sub-additivity. So $S \in \mathcal{M}$. ✓

Step 5: General countable unions. For arbitrary (not necessarily disjoint) $A_n \in \mathcal{M}$, write $\bigcup A_n$ as a disjoint union using $B_n = A_n \setminus \bigcup_{k<n} A_k \in \mathcal{M}$, and apply Step 4.

Part (b): μ is a complete measure.

Countable additivity follows from (7) with $E = S$: $\mu(S) = \mu^*(S) = \sum \mu^*(A_k) = \sum \mu(A_k)$.

Completeness: if $\mu^*(N) = 0$ and $A \subseteq N$, then for any E: $\mu^*(E \cap A) \leq \mu^*(A) \leq \mu^*(N) = 0$, so $\mu^*(E \cap A) + \mu^*(E \cap A^c) = \mu^*(E \cap A^c) \leq \mu^*(E)$. The reverse holds by monotonicity ($E \cap A^c \subseteq E$). Wait, actually we need $\mu^*(E \cap A^c) \geq \mu^*(E) - \mu^*(E \cap A) = \mu^*(E)$ and $\mu^*(E \cap A^c) \leq \mu^*(E)$. These give equality. So $A \in \mathcal{M}$.

Uniqueness under σ-finiteness: follows from Corollary **??** applied to the π-system $\mathcal{A}$, after reducing to the finite-measure case using the σ-finite decomposition. □

4.4 Application: Lebesgue Measure

Theorem 4.3 (Existence of Lebesgue Measure)**.** *There exists a unique measure λ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ such that $\lambda((a, b]) = b - a$ for all $a < b$. This measure is σ-finite and translation-invariant: $\lambda(A + x) = \lambda(A)$ for all $A \in \mathcal{B}(\mathbb{R})$ and $x \in \mathbb{R}$.*

Proof sketch. Let $\mathcal{A}$ be the algebra of finite disjoint unions of half-open intervals $(a_i, b_i]$. Define $\mu_0(\bigsqcup_i (a_i, b_i]) = \sum_i (b_i - a_i)$. One verifies μ_0 is a pre-measure on $\mathcal{A}$ (the key step is showing countable additivity, which uses the Heine–Borel theorem: if $(a, b]$ is covered by countably many $(a_n, b_n]$, then $b - a \leq \sum (b_n - a_n)$).

By Carathéodory's theorem (Theorem 4.2), μ_0 extends uniquely to $\sigma(\mathcal{A}) = \mathcal{B}(\mathbb{R})$ since μ_0 is σ-finite ($\mathbb{R} = \bigcup_{n=1}^\infty (-n, n]$ with $\mu_0((-n, n]) = 2n < \infty$). □

Theorem 4.4 (Non-Measurable Sets (Vitali))**.** *There exists a set $V \subseteq [0, 1]$ that is not Lebesgue measurable.*

Proof. Define an equivalence relation on $[0,1)$: $x \sim y$ iff $x - y \in \mathbb{Q}$. By the Axiom of Choice, select one representative from each equivalence class; call this set V.

For $q \in \mathbb{Q} \cap [0,1)$, define $V_q = \{x + q \mod 1 : x \in V\}$ (shift V by q modulo 1). The V_q are disjoint (if $v_1 + q_1 \equiv v_2 + q_2$ then $v_1 - v_2 \in \mathbb{Q}$, so $v_1 \sim v_2$, so $v_1 = v_2$ and $q_1 = q_2$). Also $[0,1) = \bigsqcup_{q \in \mathbb{Q} \cap [0,1)} V_q$ (every x is equivalent to some $v \in V$, and $x = v + q$ for some rational q).

If V were measurable, then each V_q would have the same measure (by translation invariance of Lebesgue measure modulo 1). Call this c.

By countable additivity: $1 = \lambda([0,1)) = \sum_q c$. But $\sum c$ is either 0 (if $c = 0$) or ∞ (if $c > 0$) — never 1. Contradiction. $\square$

Remark 4.3. Vitali's theorem shows why we cannot define Lebesgue measure on $\mathcal{P}(\mathbb{R})$: not all subsets of $\mathbb{R}$ are measurable. This is the fundamental reason for restricting to σ-algebras. In probability, this subtlety is rarely encountered (finite MDPs use $\mathcal{P}(\mathcal{S})$), but it justifies the formalism and is important for continuous state/action spaces.